



Hot Chips Tutorial, Part-IV

RISC-V in Automotive: Opportunities and Challenges

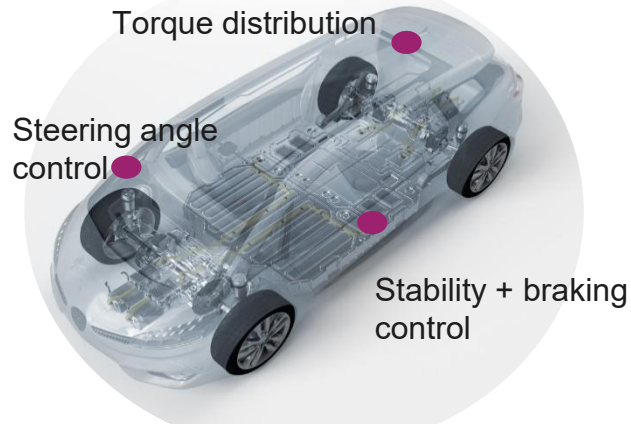
T B Roecker



Microcontroller compute requirements: Combine application-specific and cross-domain aspects

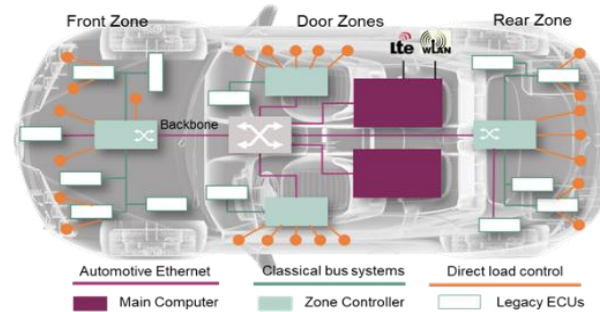
Application

Chassis/Powertrain



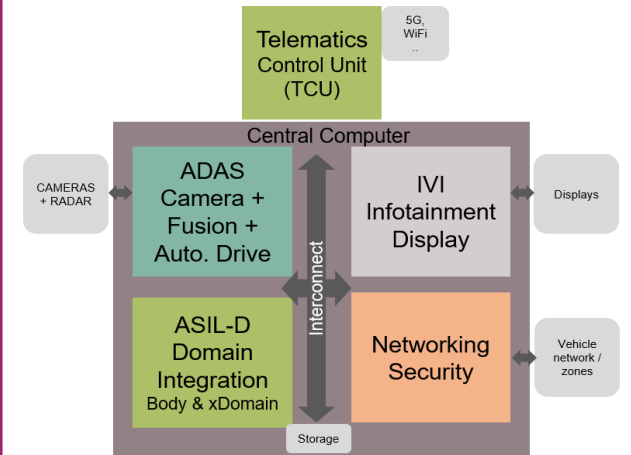
Real-Time, low latencies,
highest Integrity-Level

Zone Controllers



Power-efficiency, Secure
Compute, Integration capability

ADAS



Memory-bandwidth, Vector-
Performance

X-Domain

Safety



Security

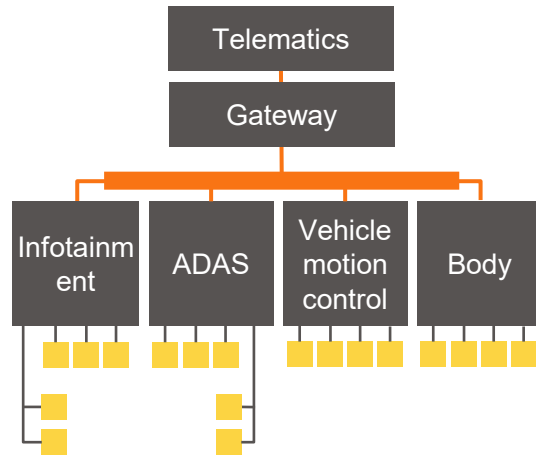
Availability



Analyzability

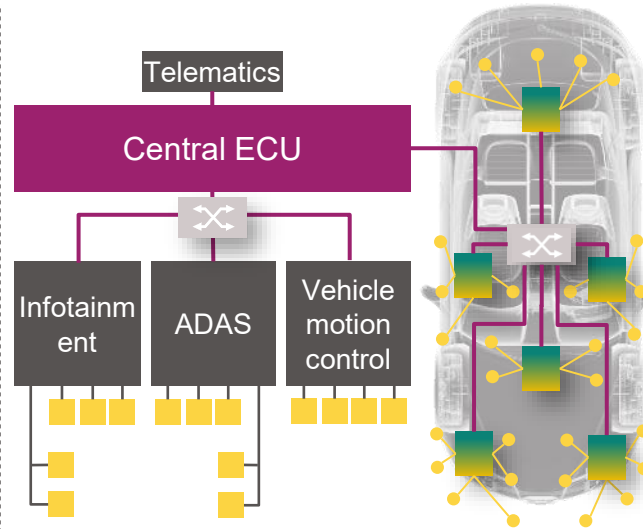
Architectural transformation of E/E-architecture drives functional consolidation → Zone Control Units (ZCU)

Domain architecture

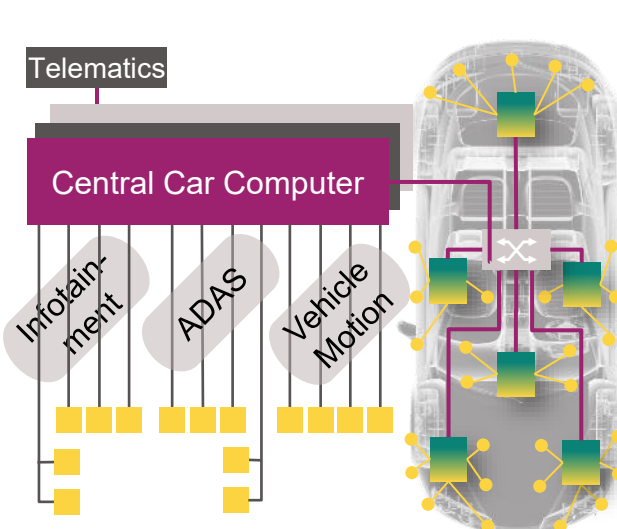


- › Central Gateway
- › Powerful domain controllers
- › Separate function domains

Mixed Domain/Zone Architecture

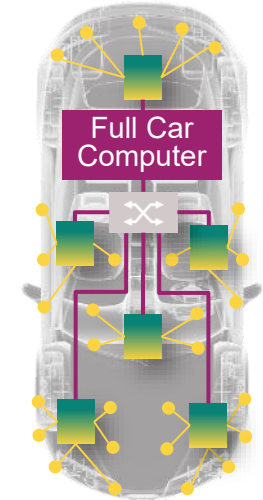


- › ADAS, Infotainment, Vehicle motion (powertrain, chassis) remain in own, separated domains
- › **One central ECU mainly controls Body-zones:**
 - Direct control of local body loads and their diagnostics
 - In charge of power distribution & management



- › Former domain controllers **evolve** to mighty **central car computers**
- › **More complex zones**
 - Comprising body and comfort
 - Partly vehicle motion functions
 - Partly ADAS functions
 - Sharing of wiring leads to a function agnostic network
 - Increased saving of weight/cost

Full Car Computer



- › Single or multiple Central Car computer(s)
- › **Most advanced zones**
- › Improved sharing of wiring
- › full function agnostic network
- › Lowest weight and cost of wiring harness

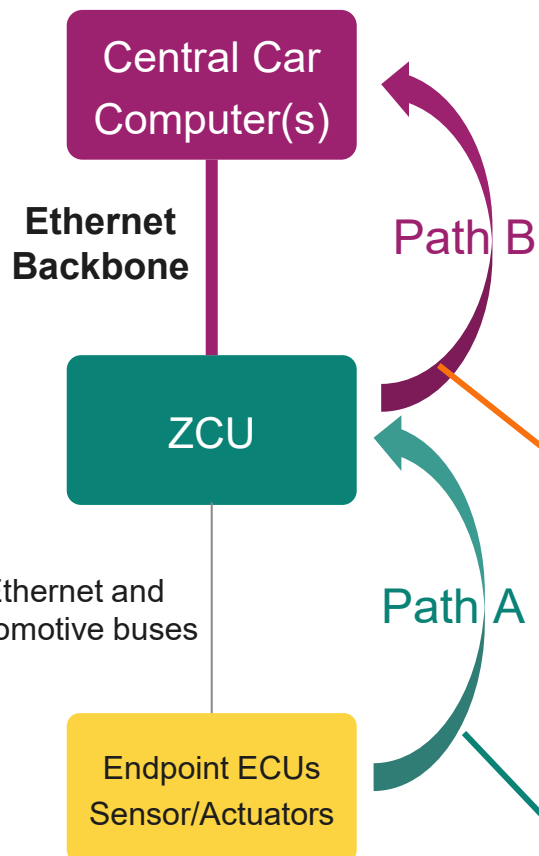
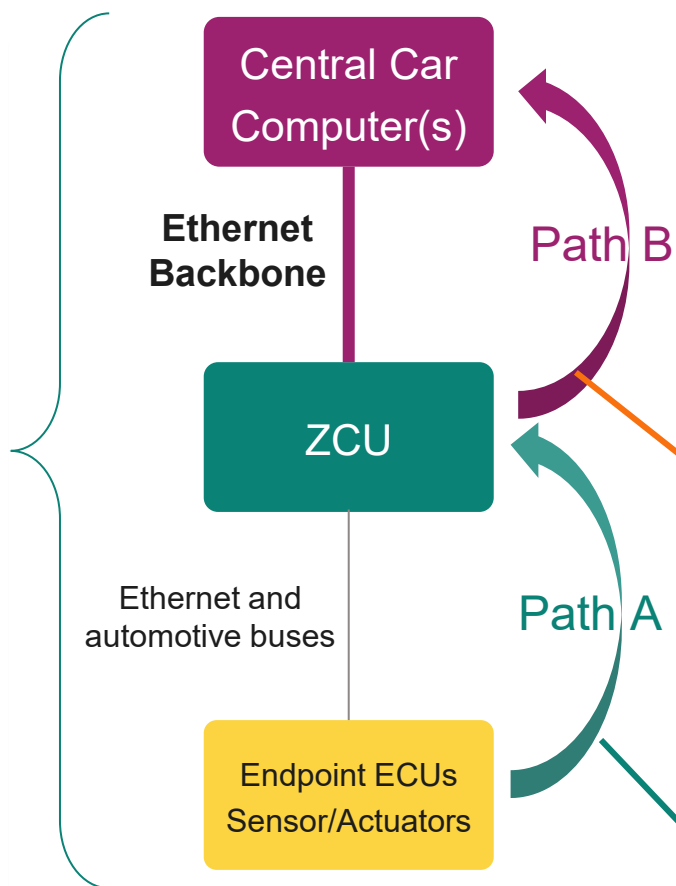
E/E-architecture: Real-time & time bounded characteristics

Properties	Real-time fast control	Real-time zone & domain	Performance time-bounded	High-performance compute
Deadline (e.g., FTTI)	$\leq 10\text{ms}$	10ms	100...500 ms	Multiple seconds
Task length	50 μs (e.g., E-motor control)	500 μs (e.g., RT COM)	200 ms (e.g., control loop)	No task, data frame-rate driven
Interrupt latency	50 ns - 100 ns	10 μs	500 μs	10 ms
WCET* calculation possible?	Yes	Yes	Yes	No, optimization for average performance
	Direct control loops (sense-control-actuate)	Tactical functions (vehicle motion control)	Strategic functions (vehicle coordination)	Strategic functions (vehicle management)

*WCET: worst case execution time

Consolidation: Different options for centralized E/E-architectures

- Automotive E/E-architecture → Comprehensive design and organization of wiring, sensors, actuators, control units, and software
- Integration scenarios for ZCU in central car E/E architectures:
 - Path A: Optimized multi domain zone controllers
 - Path B: Low-complexity I/O aggregators



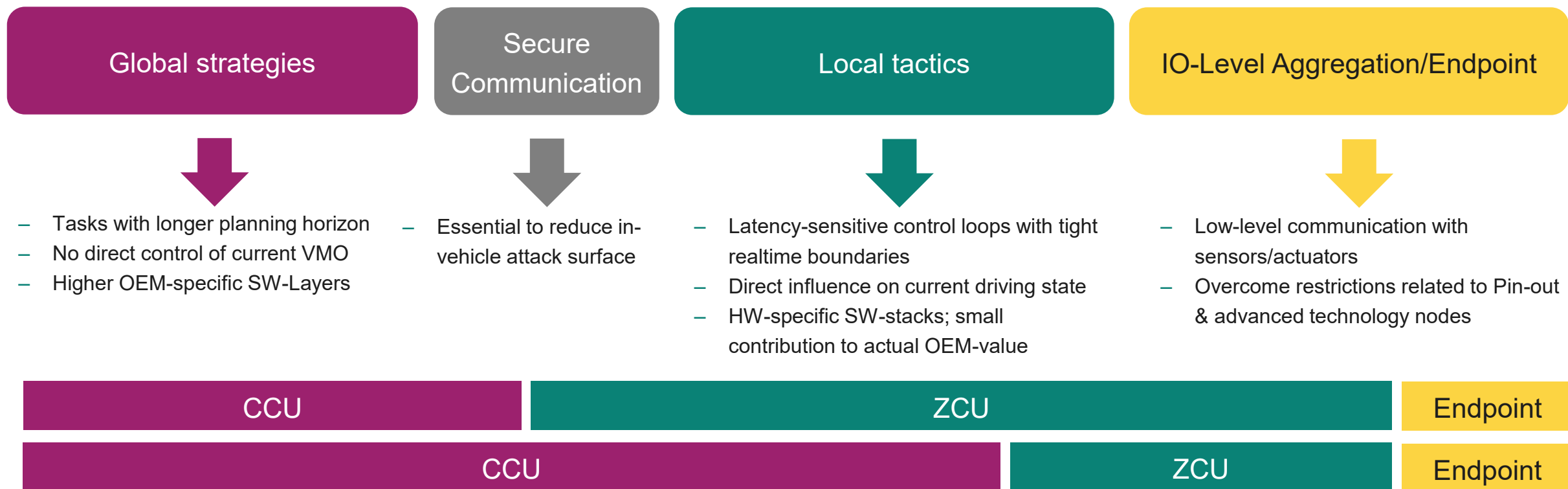
Path B

- Central car compute(s) fully support domains functions
- Zone controllers are connecting end-points and central car computer(s)
- **No intelligence in zone controllers**

Path A

- Zone controllers partially consolidate end-points ECUs
- Zone controllers support power distribution
- Zone controllers support local HW-related tactics
- **Zone controllers are connecting end-points and central car computer(s)**

Discussion of consolidation options → Option A advantageous



Additional requirements:

- Overhead related to data-routing should be kept minimal
- Qualification efforts for Realtime functions need to be minimized
- Multi-supplier strategy for end-points demand limiting impact in case of replacement
- Power-distribution follows trend of decentralization

**Option A
advantageous**

Reasons

- Physical partitioning follows functional segmentation → Better scalability SW and HW, easier maintenance, reduced routing
- System cost reduction with consolidation of end-point ECUs and power distribution in the zones
- Clear separation between OEM SW with added value and HW-specific SW

Zonal Use-Cases drive heterogenous compute requirements



Control of local Sense/Actuate layer → Realtime-Workloads



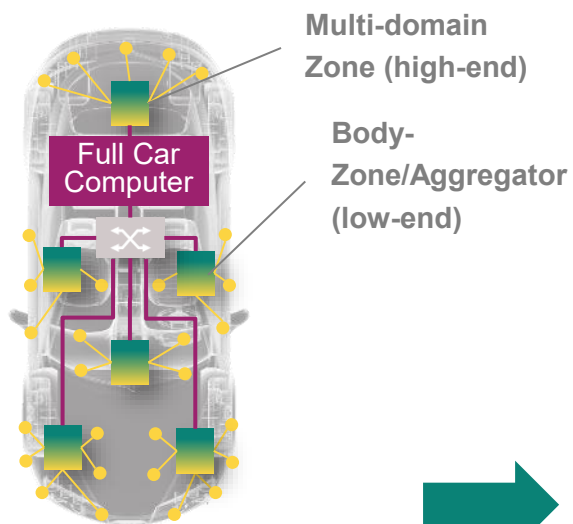
IO-Aggregation and abstraction & Prefusion-Scenarios



Power-distr. & - monitoring, Smart Fuses, predictive maintenance



Keyword-recognition, Siren-detection, Media-Playback



- ZCU integrates workloads with different compute requirements: Realtime, DSP, AI-Inference, Low-Power, Service-Layer etc
- Scalability required for different ZCUs positioned differently in the car: Multi-Domain vs Body-Zones
- Safety- and Security are a must-have, yet also here scalability is key

➔ Compute-Architecture needs to be scalable, deployable, still assuring good PPA

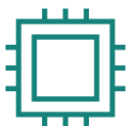
Heterogeneity is driven further by Super-Integration scenarios



- Deploy non-RT critical workloads to MPU-island brings benefits in terms of ecosystem, competence management, migration scenarios



- Implementation of central body and/or networking functions that better fit on a POSIX / adaptive ASR software stack



- Efficient external memory utilization for applications dealing with very high data amounts like data logging

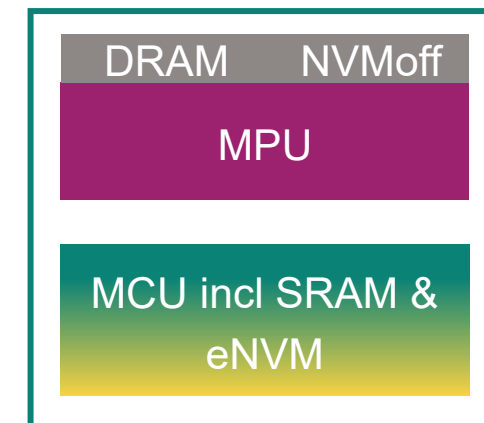


- Freedom from interference for limited number of safety relevant adaptive applications (e.g. ADAS or vehicle motion); this means better scalability from L2 towards L2+/L3



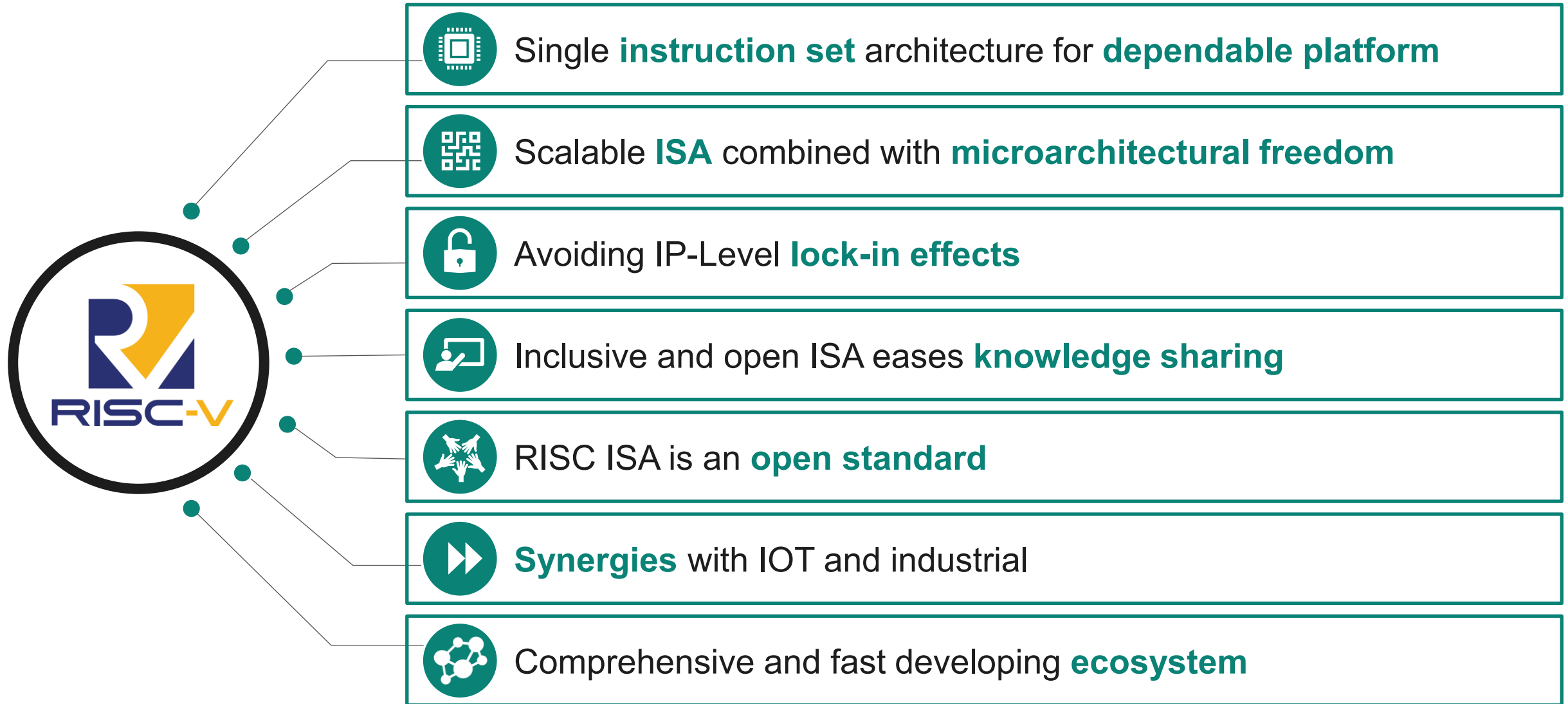
- Providing faster boot and wake-up of optimized adaptive stack and critical services compared to SoC startup times

Super-Integration ZCU



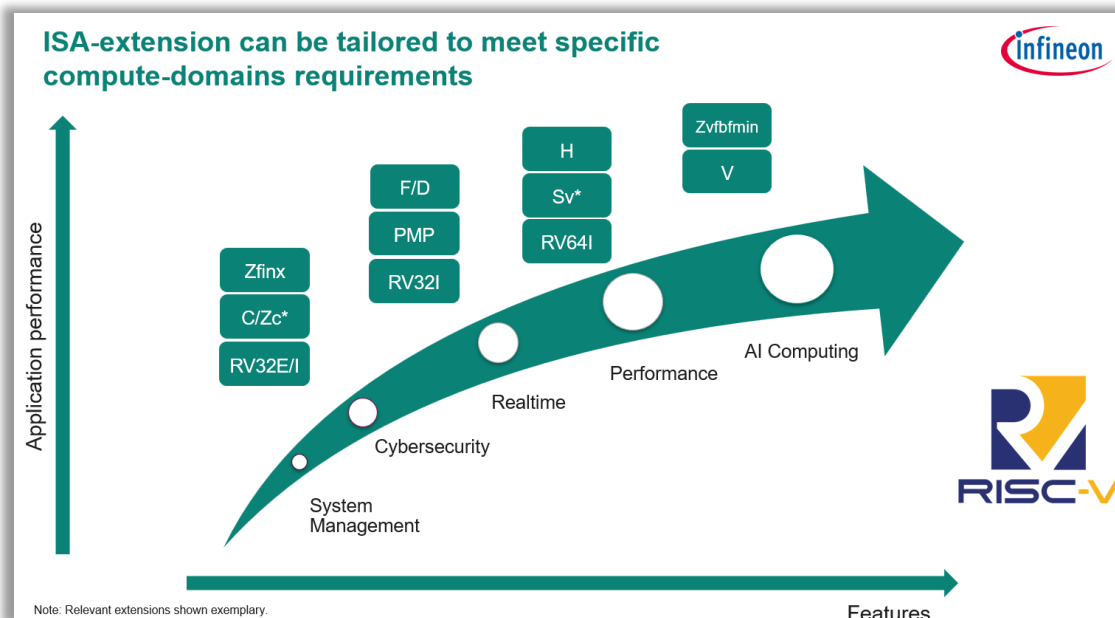
Super-integration enables further cost-savings down E/E-hierarchy

RISC-V as ISA in automotive applications and beyond offers great opportunities



Unified profiles need to cover automotive requirements

RVI AUTO-SIG discussion



RISC-V Automotive Profile(s)



RVA23 - MMU-Based Domain / Zonal, IVI, Central Compute

- Agreed in our last call to keep RVA23 common for auto/non-auto to reduce fragmentation.

RVM - MMU-Less Domain / Zonal, ECU, Sensor, Actuator

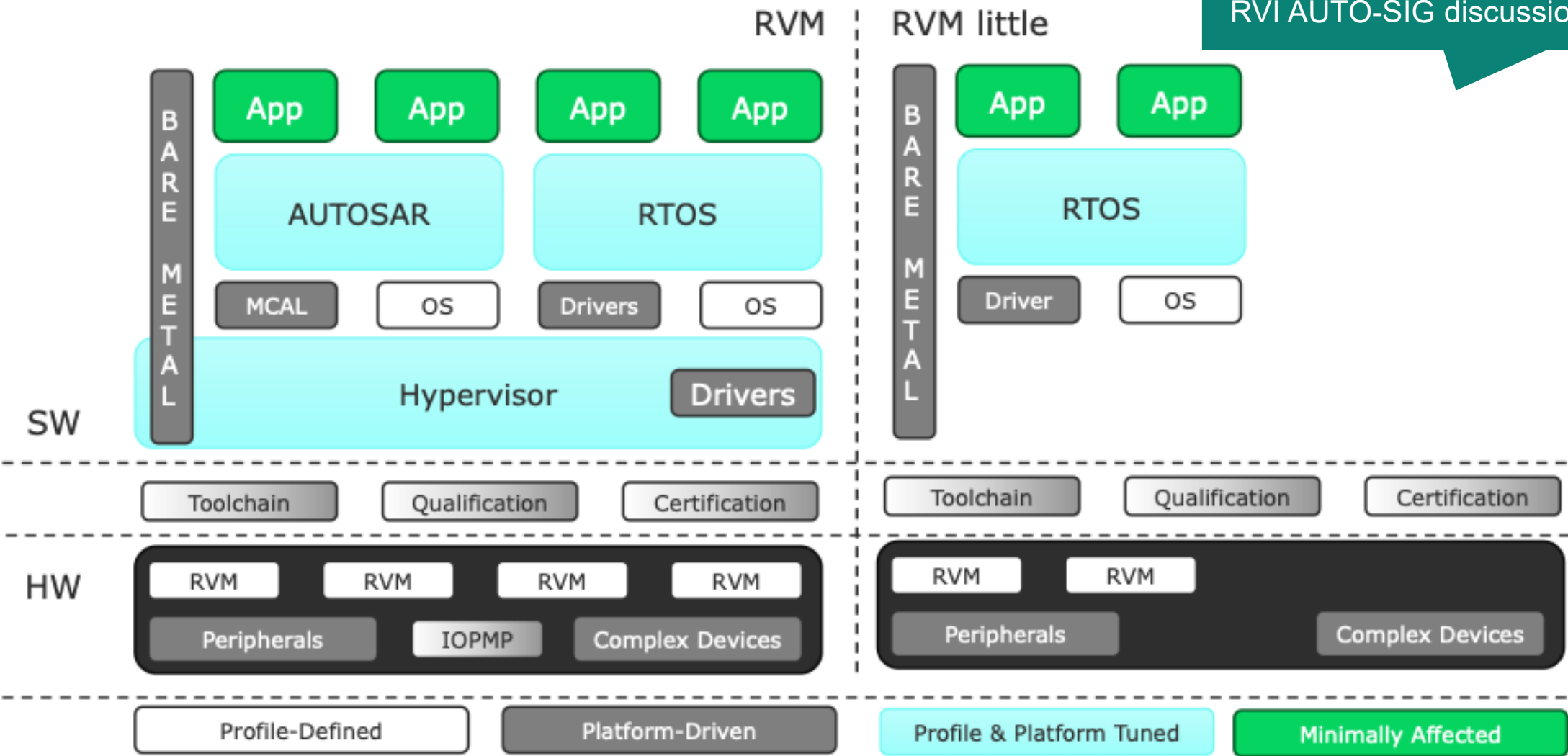
- Proposal
 - Keep RVM common for auto/non-auto to reduce fragmentation.
 - Must have a single profile and specification - If we don't, they WILL diverge
 - But the wide range of device types/price makes optionality essential

- Distinction is not so much about AUTO/Non-AUTO, but definition for industry-grade, safety-/security-related product to address needs of revised E/E-architecture → Clearly requires class-based approach/optionalities
 - Start with base comprised of RV32E/I & M+U, PMPs, RNMI, A, B, Cache-Mmgt, C/Zc*.
 - Add optionalities: Supervisor (Svbare), Float, Vectors, H-Support, CFI, AIA, ACLIC
- Higher abstraction levels mandate for less specifics → Missing RVM as major gap/focus first
 - CCU → RVA23 and successors, ZCU → 'large' RVM, Endpoint → 'little' RVM or even SW-less nodes
- Aspects apply even more to subsystem/compute-platform definition (IRQ, D/T, Mixed criticality, Fault structure)

SW-Stacks to be considered for scalability of RVM, eg need for virtualization solution



RVI AUTO-SIG discussion

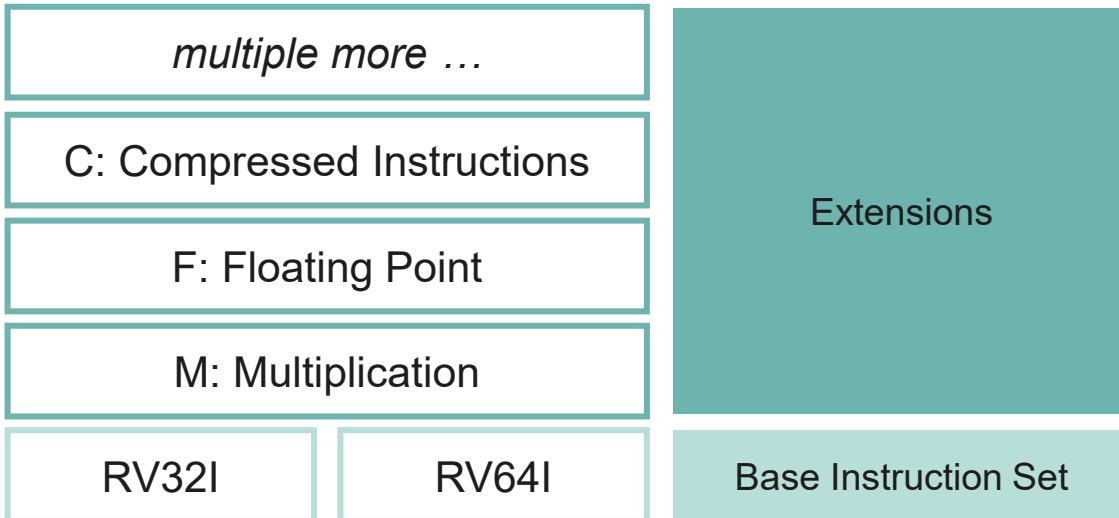


Profiles are important but RISC-V specifies the ISA, not implementation specifics of a CPU

Part of ISA Specification



- › Functional specification of behavior and corresponding hardware-software interface
- › Modular granularity via extensions



Not part of ISA Specification



- › Implementation specific details
 - › Memory hierarchy
 - › Latency of instructions
 - › Latency to memories
 - › Nb. And type of interfaces
 - › Microarchitecture
 - › ...
- › Functional safety aspects
 - › Lock-step
 - › Error tracking and correction
 - › ...
- › Cluster parts
 - › Cache coherency
 - › ...

From opportunities to challenges for RV in Automotive

„Detailed aspects of the ISA mostly don't matter, more important is microarchitectural optimization“



ISA

Requirements for sourcing



„MCUs major attributes are memory, Connectivity options, specific HW, and finally BOM-costs“

„Availability of auto-qualified toolchains is deciding on success“



Speed of standardization

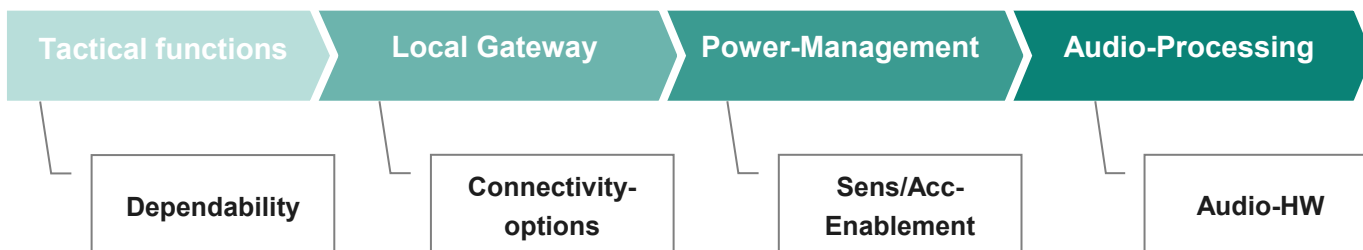
Scalability & Migration



„Value-chain targets reduction of E/E-architectures TCO by scalability. Migration path remains critical.“

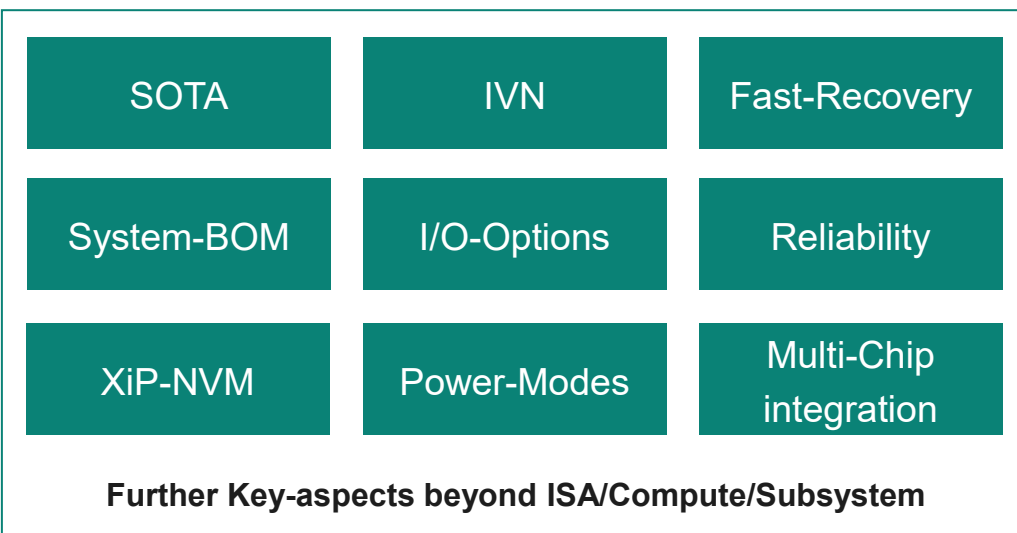


What does this mean for the role of RISC-V in E/E-architectures?



RISC-V can be **accelerating technology** for ZCUs.
Application- and cost-fit are essentially determined by
factors beyond ISA/IP-Level.
SDV amplifies focus on **system-solution strategy**

However, form-fit-function is notably determined by SoC-architecture, Ecosystem-Enablement, Ease of Deployment



What does this mean for RISC-V IP?

- Long-term reliable automotive offering essential
- Core-Portfolio requires wide scalability
- IP needs to be optimized for automotive implementations
- Acceleration along value-chain will only happen based on standardized HSI & toolchain support



